Merkle Tree

BY CODE EATER

Contents

Hashing

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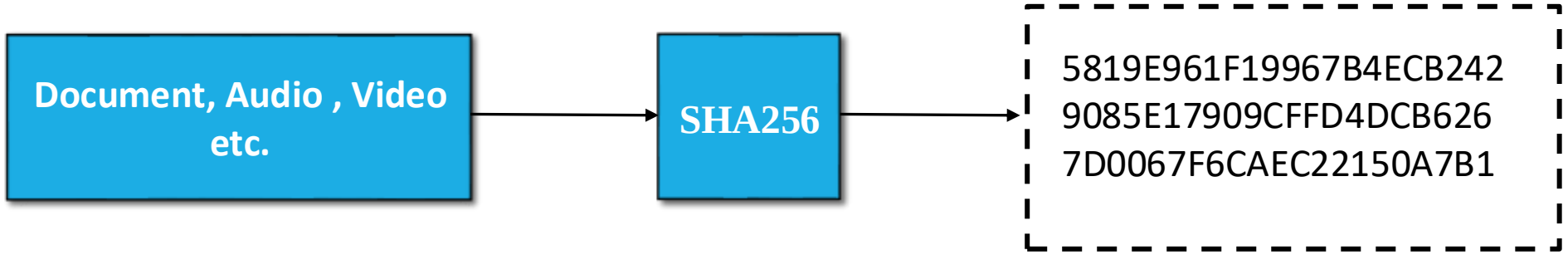
Hashing

Hashing Algorithm



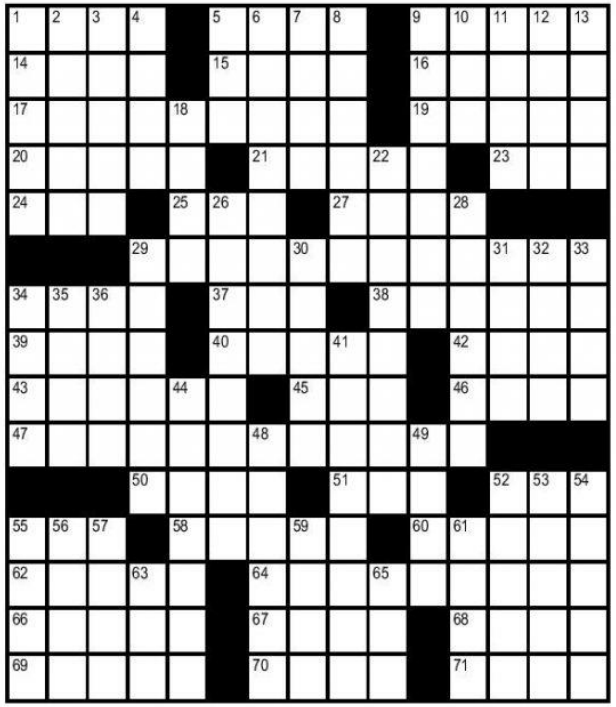
This has **64 hexadecimal characters.**
Each character is of **4 bits.**
So in total it has 64* 4 bits i.e. **256 bits**.

Crossword Puzzle

WORCESTER SUN CROSSWORD

BY FRANK VIRZI

Hidden Strength

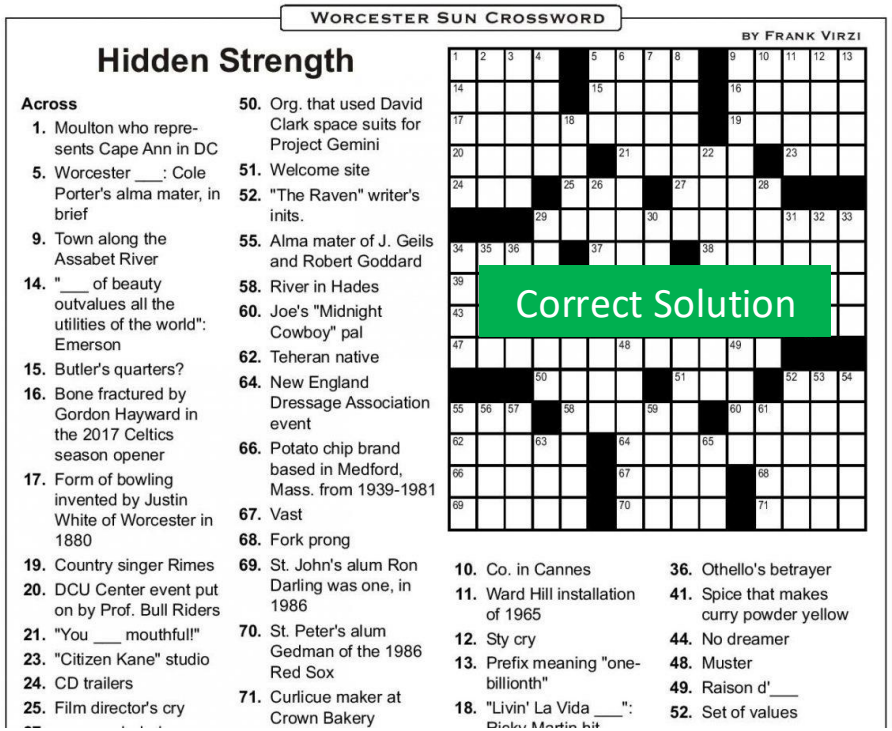


Across

1. Moulton who represents Cape Ann in DC
5. Worcester ___: Cole Porter's alma mater, in brief
9. Town along the Assabet River
14. "___ of beauty outvalues all the utilities of the world": Emerson
15. Butler's quarters?
16. Bone fractured by Gordon Hayward in the 2017 Celtics season opener
17. Form of bowling invented by Justin White of Worcester in 1880
19. Country singer Rimes
20. DCU Center event put on by Prof. Bull Riders
21. "You ___ mouthful!"
23. "Citizen Kane" studio
24. CD trailers
25. Film director's cry
50. Org. that used David Clark space suits for Project Gemini
51. Welcome site
52. "The Raven" writer's inits.
55. Alma mater of J. Geils and Robert Goddard
58. River in Hades
60. Joe's "Midnight Cowboy" pal
62. Teheran native
64. New England Dressage Association event
66. Potato chip brand based in Medford, Mass. from 1939-1981
67. Vast
68. Fork prong
69. St. John's alum Ron Darling was one, in 1986
70. St. Peter's alum Gedman of the 1986 Red Sox
71. Curlicue maker at Crown Bakery

10. Co. in Cannes
11. Ward Hill installation of 1965
12. Sty cry
13. Prefix meaning "one-billionth"
18. "Livin' La Vida ___": Ricky Martin hit
36. Othello's betrayer
41. Spice that makes curry powder yellow
44. No dreamer
48. Muster
49. Raison d'___
52. Set of values

Crossword Puzzle



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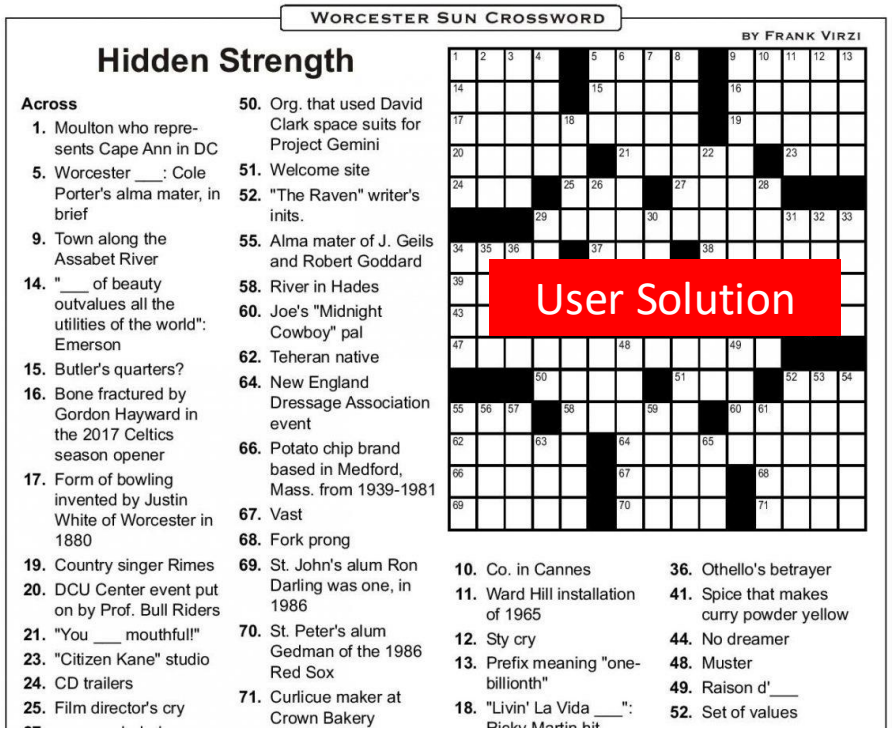
Correct Solution

Hash

5819E961F19967B4ECB242
9085E17909CFFD4DCB626
7D0067F6CAEC22150A7B1

Crossword Puzzle



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User Solution

Hash

18ee24150dcb1d96752a4d
6dd0f20dfd8ba8c38527e4
0aa8509b7adecf78f9c6

Crossword Puzzle

5819E961F19967B4ECB242
9085E17909CFFD4DCB626
7D0067F6CAEC22150A7B1

Correct Solution

≠

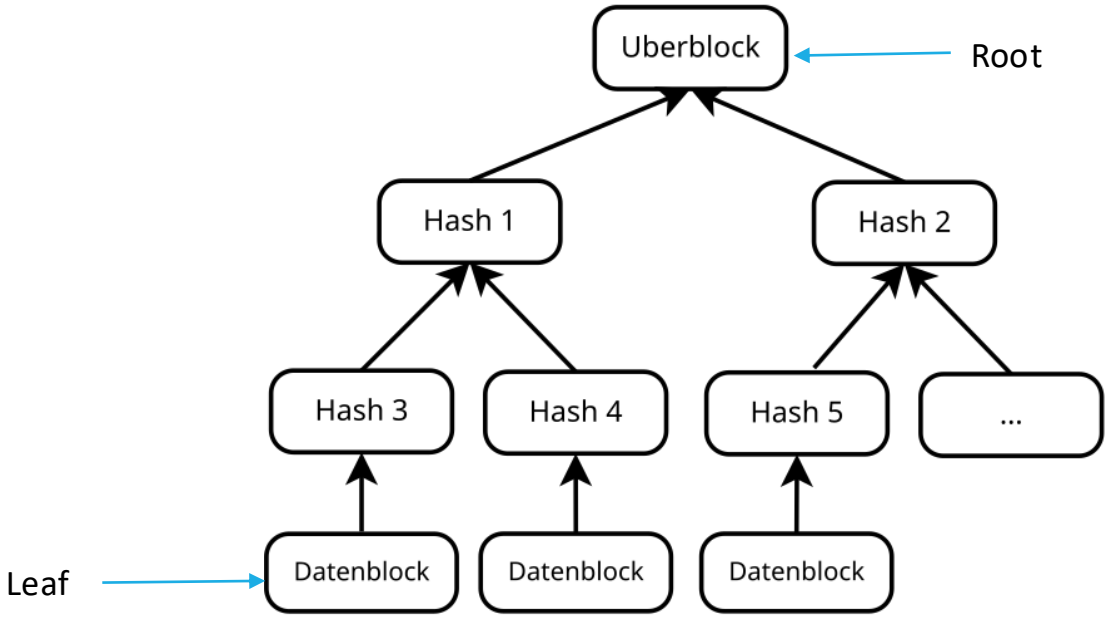
18ee24150dcb1d96752a4d
6dd0f20dfd8ba8c38527e4
0aa8509b7adecf78f9c6

User Solution

Merkle Tree

Merkle Tree

A Merkle tree is a data structure that uses hashes to verify the authenticity of data



Source:wikimedia.org

H1,H2,H3,H4 are hash value of D1,D2,D3,D4.

H1,H2,H3,H4 are hash value of data D1,D2,D3,D4.

H1

H2

H3

H4

H1,H2,H3,H4 are hash value of data D1,D2,D3,D4.

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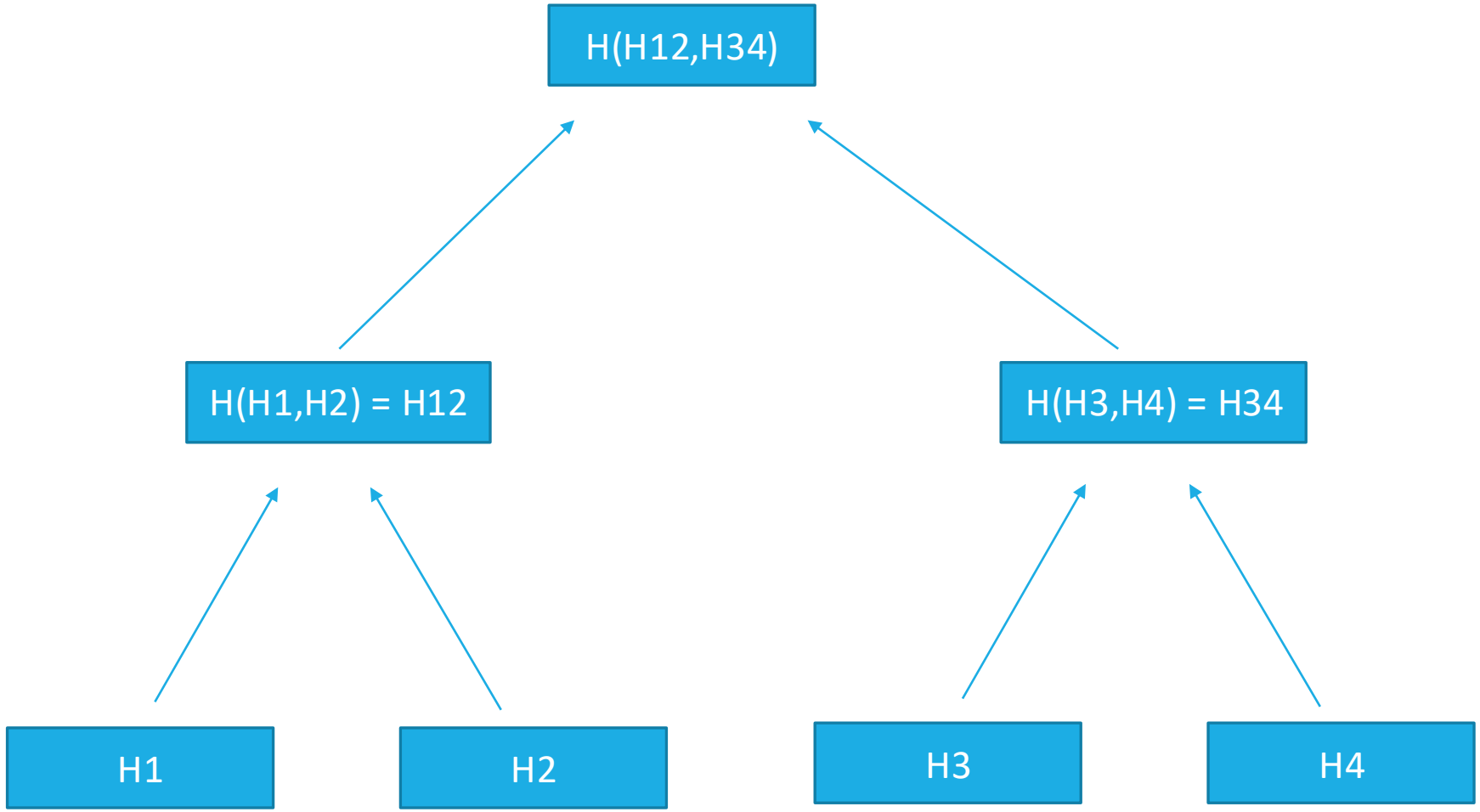
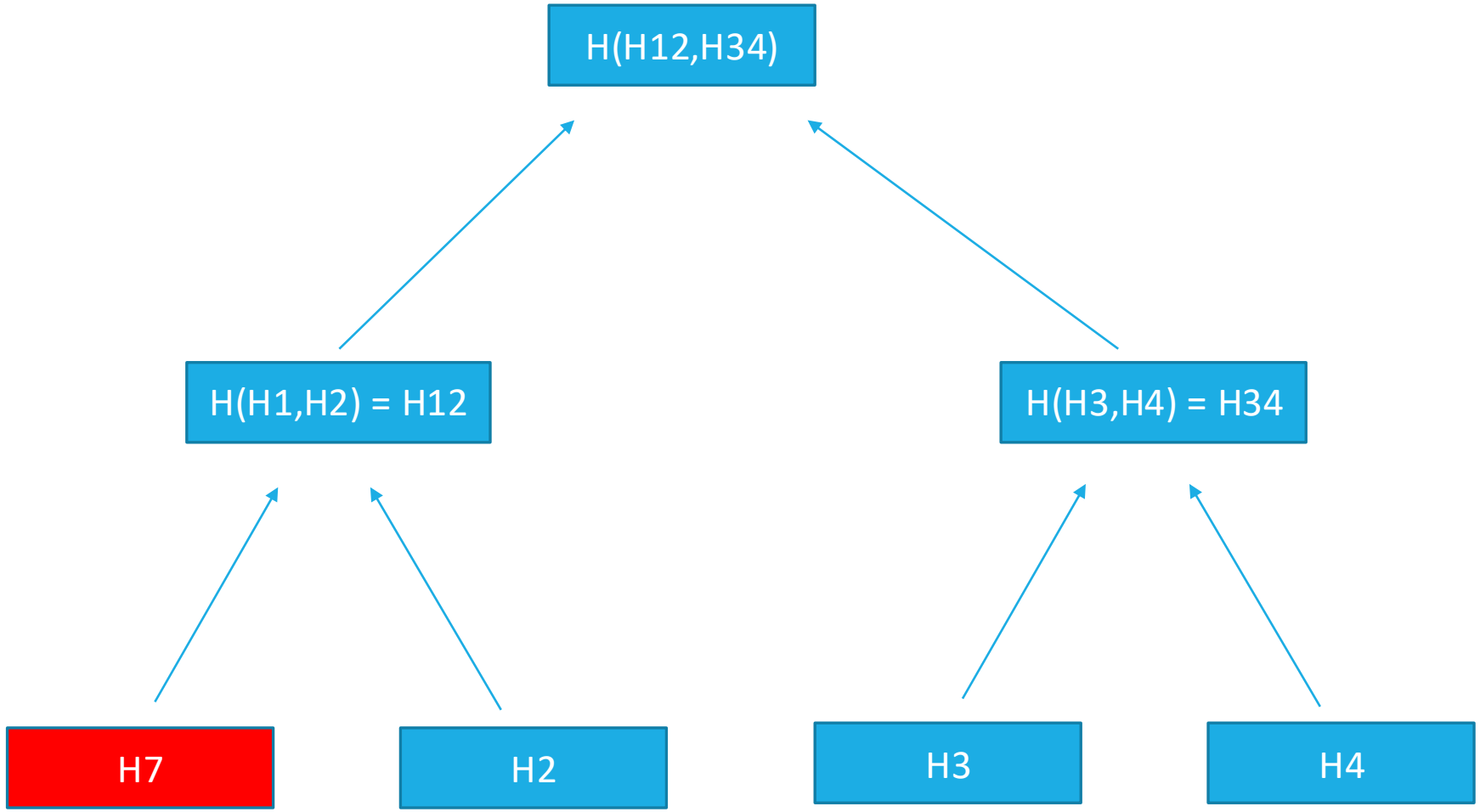


Merke Tree Simulator

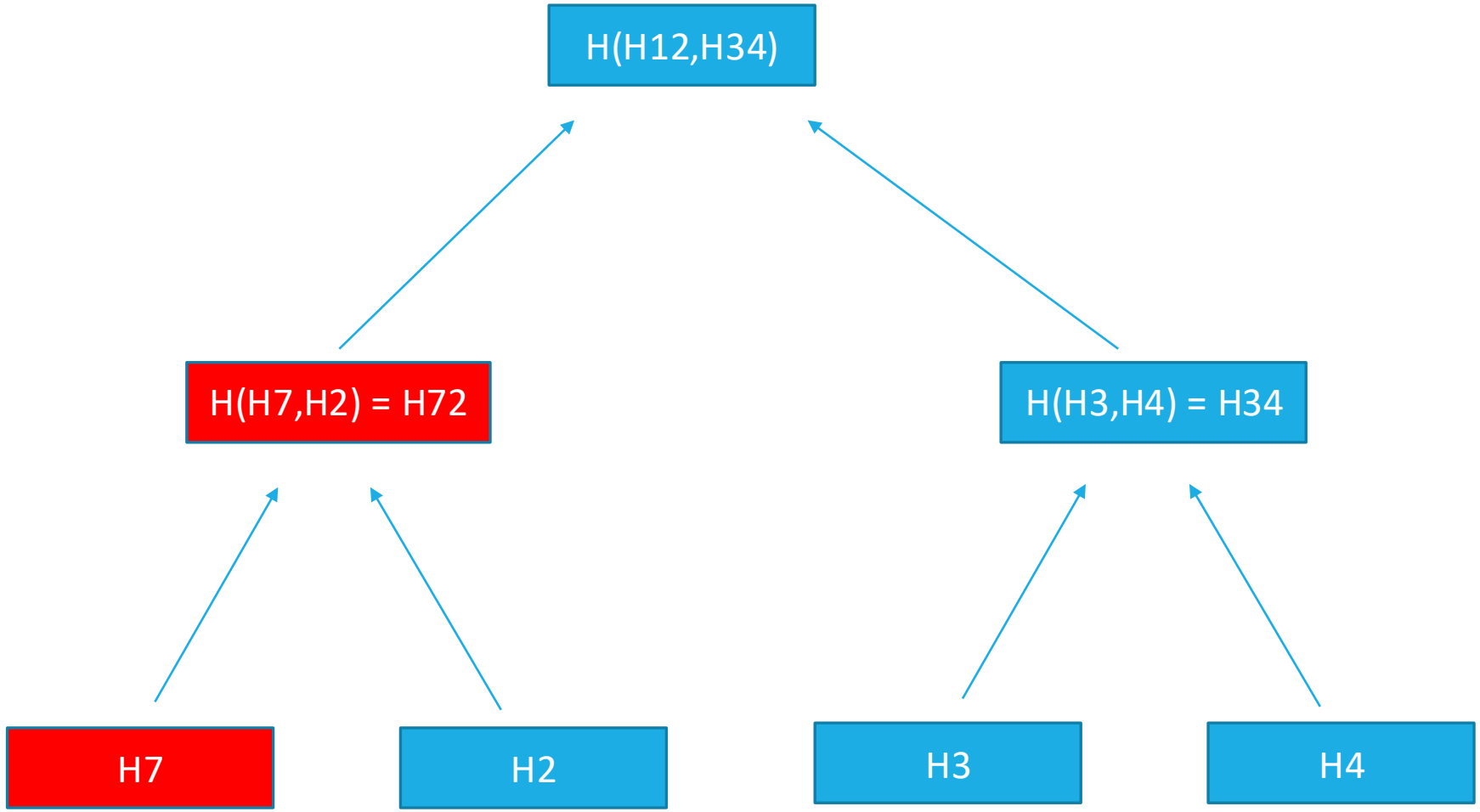
Merkle Tree Simulator – Click Here

Tempering Of Data

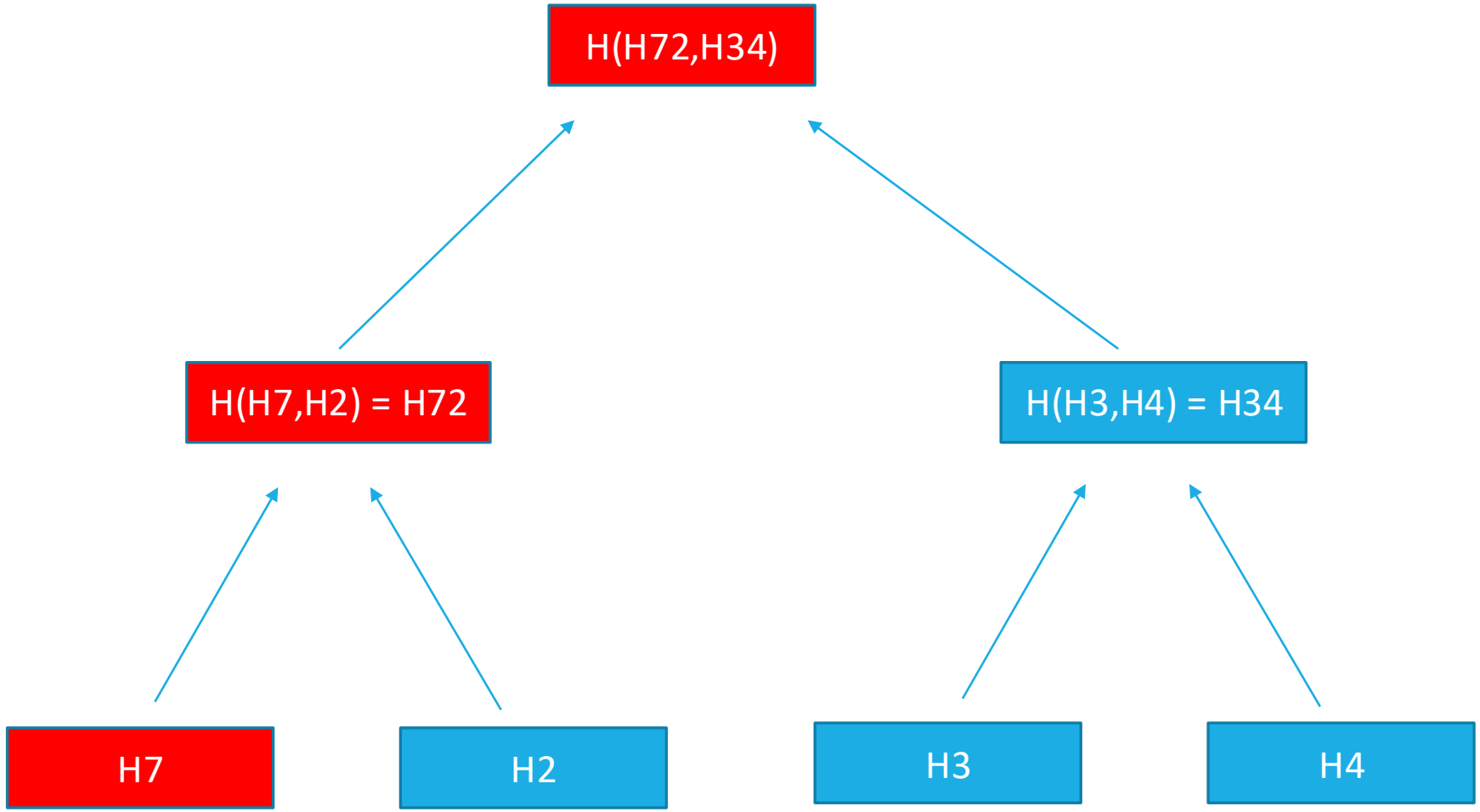
H1,H2,H3,H4 are hash value of T1,T2,T3,T4.



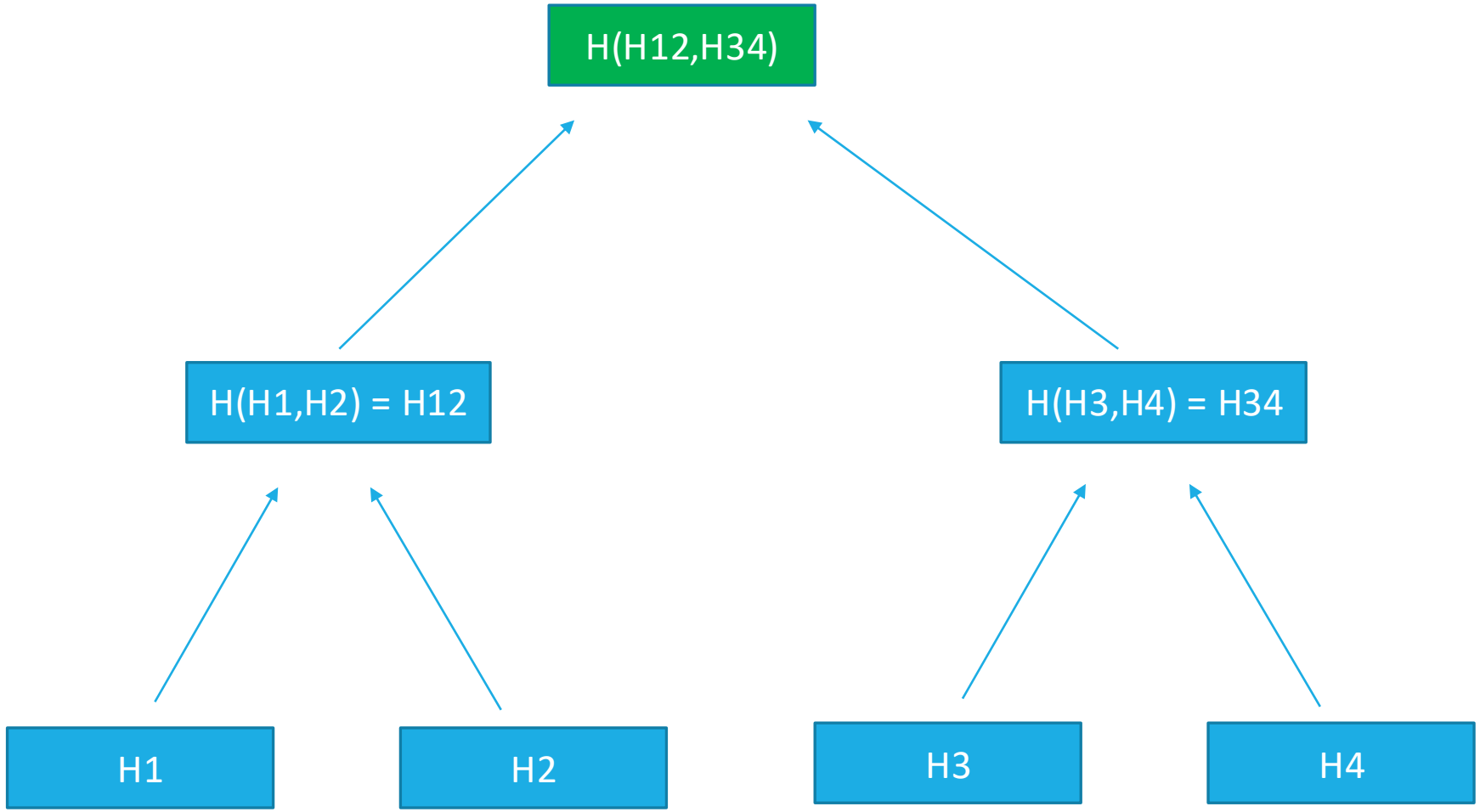
H1,H2,H3,H4 are hash value of T1,T2,T3,T4.



H1,H2,H3,H4 are hash value of T1,T2,T3,T4.



H1,H2,H3,H4 are hash value of T1,T2,T3,T4.



Merkle Proof

A Merkle proof is a cryptographic method to verify that a specific data piece belongs within a large dataset.

Purpose - Using a Merkle tree, it efficiently confirms membership by tracing a path from the data to the root hash, without needing to check the entire dataset.

Merkle Tree Proof Generator

Merkle Tree Proof Generator Code – Click Here

Thank You